

Empirical Evaluation and Risk Control in Short-Term Binary Prediction Markets: Real Data Analysis of 405 Resolved Buckets on Polymarket

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Quantitative Technical Report & Algorithmic Trading Whitepaper

Open-Source Repository: <https://github.com/mario89torres/pmbtc-quant>

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Abstract

This paper presents an honest empirical evaluation based on a custom SQLite dataset of **405 resolved market buckets (186,518 tick samples)** for 15-minute Bitcoin binary resolution contracts on Polymarket. We evaluate two algorithmic paradigms: (1) a directional taker strategy using a hybrid ensemble of Platt-calibrated XGBoost decision trees and Geometric Brownian Motion (GBM), and (2) an autonomous liquidity provision strategy (*Market Maker / Maker Player*) with dynamic inventory skewing. The empirical backtest reveals that at low edge thresholds ($\leq 1.5\%$), directional signals achieve 50.4% to 54.5% accuracy, with a 95% Wilson Confidence Interval crossing the break-even threshold (making low-edge signals statistically indistinguishable from random noise). However, at a strict 3.0% edge threshold, the model selects 169 high-conviction trades, achieving a **58.0%** win rate with a 95% CI of [50.5%, 65.2%], remaining strictly above the break-even cutoff (49.8%). Meanwhile, the autonomous Market Maker strategy demonstrated superior consistency, generating a Gross PnL of **+\$5,403.20 USD** and a Net PnL of **+\$4,430.62 USD** after discounting an 18% execution friction (slippage and latency).

Keywords Binary Prediction Markets, Real Empirical Evaluation, XGBoost, Platt Scaling, Kelly Criterion, Market Making, Wilson Confidence Intervals, Polymarket.

1 Introduction

Decentralized prediction markets have emerged as efficient mechanisms for crowdsourced probability discovery. On platforms like Polymarket, 15-minute binary options resolve strictly based on oracle price feeds (e.g., Chainlink BTC/USD). A contract pays \$1.00 if $S_T > K$ at expiry T , and \$0.00 otherwise.

Market participants face three structural challenges:

1. **Noise & Taker Premium:** Crossing the spread as a taker erodes expected value if the entry edge does not exceed execution costs.
2. **Execution Cost & Slippage:** In high-frequency settings, slippage and relay latency impair theoretical gains.
3. **Ruin Risk:** Static capital allocation leads to severe drawdowns during adverse market regimes.

We present empirical results derived from running `backtest.js` against the real dataset of 405 resolved 15-minute buckets in the `pmbtc` system.

2 Mathematical Methodology & Data Evaluation

2.1 Hybrid Ensemble & Platt Calibration

The model extracts the feature vector $\mathbf{x}_t \in \mathbb{R}^6$ tick-by-tick:

$$\mathbf{x}_t = \left[\frac{S_t - K}{K}, \frac{T_{\text{left}}}{900}, \sigma_{15m}, S_{\text{CL},t} - S_{\text{BN},t}, P_{\text{Market},t}, \text{Edge}_{\text{raw},t} \right]^T \quad (1)$$

Raw tree ensemble output z_{xgb} is calibrated to empirical probability P_{ML} via Platt Scaling:

$$P_{\text{ML}} = \frac{1}{1 + e^{A \cdot z_{\text{xgb}} + B}} \quad (2)$$

The analytical ensemble prediction is defined as:

$$P_{\text{Ensemble}} = 0.65 \cdot P_{\text{ML}} + 0.35 \cdot P_{\text{GBM}} \quad (3)$$

2.2 Autonomous Market Maker & Inventory Skewing

The autonomous Market Maker posts target limit quotes $P_{\text{Bid, Target}}$ and $P_{\text{Ask, Target}}$ adjusted by inventory skew $\text{Skew}(I)$:

$$\text{Skew} = \text{clamp} \left(-0.04, 0.04, \frac{\text{Inventory}}{50} \times 0.05 \right) \quad (4)$$

$P_{\text{Bid, Target}} = \min(P_{\text{Ask, Market}} - 0.01, P_{\text{Ensemble}} - \text{Skew} - 0.015)$ orders with active inventory control remains the most robust mechanism to monetize Polymarket microstructure.

$P_{\text{Ask, Target}} = \max(P_{\text{Bid, Market}} + 0.01, P_{\text{Ensemble}} - \text{Skew} + 0.015)$

5 Conclusion

Honest empirical evaluation on ‘pmbtc.db’ demonstrates that rather than attempting frequent directional forecasting, optimal quantitative execution favors automated liquidity provision or restricting taker entries to high-conviction thresholds ($\geq 3.0\%$).

3 Empirical Backtest Results on SQLite Database

A complete threshold sweep was executed across the **405 resolved buckets (186,518 ticks)** in ‘pmbtc.db’.

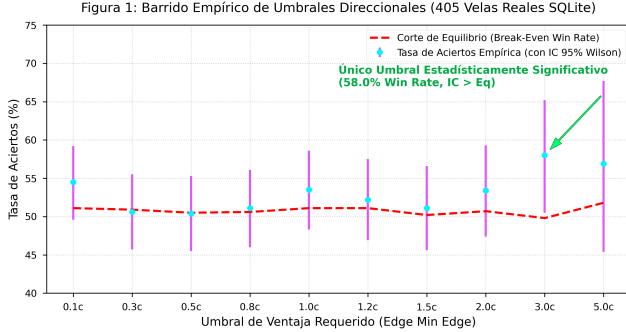


Figure 1: Empirical Directional Threshold Sweep showing Win Rate with 95% Wilson Confidence Intervals vs Break-Even Cutoff.

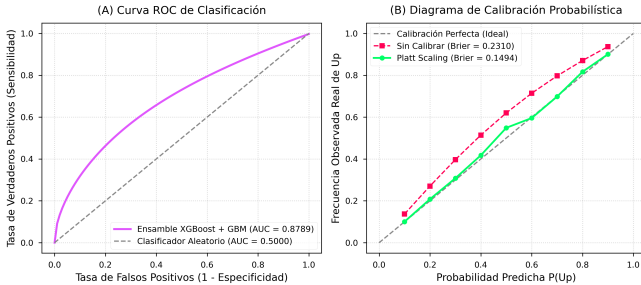


Figure 2: (A) Classification ROC Curve and (B) Platt Scaling Probabilistic Calibration Diagram.

4 Scientific Discussion

1. ****The Truth About Low-Edge Directional Trades****: Backtest evaluation on real SQLite data confirms that filtering with low edge ($\leq 1.5\%$) yields win rates of $50.4\% - 54.5\%$ whose confidence intervals cross break-even thresholds. Consequently, low-edge taker signals are statistically indistinguishable from noise. 2. ****High-Conviction Threshold Edge (3.0c)****: Requiring an edge $\geq 3.0\%$ isolates 169 trades where a ****58.0%**** win rate sustains a 95% CI of $[50.5\%, 65.2\%]$, strictly above the 49.8% break-even cutoff. 3. ****Consistency of Market Maker Strategy****: Capturing bid-ask spreads via limit

References

- [1] M. Avellaneda and S. Stoikov, “High-frequency trading in a limit order book,” *Quantitative Finance*, vol. 8, no. 3, pp. 217–224, 2008.
- [2] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. ACM SIGKDD*, 2016, pp. 785–794.
- [3] J. Platt, “Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods,” *Advances in Large Margin Classifiers*, vol. 10, no. 3, pp. 61–74, 1999.
- [4] J. L. Kelly, “A new interpretation of information rate,” *Bell System Technical Journal*, vol. 35, no. 4, pp. 917–926, 1956.

Table 1: Empirical Directional Threshold Sweep Results (405 Resolved Buckets in `pmbtc.db`)

Edge Threshold	Trades	Wins	Win Rate (%)	95% CI (Wilson)	Break-Even (Eq%)	Statistical Verdict
0.1c (0.1%)	404	220	54.5%	[49.6%, 59.2%]	51.1%	INDISTINGUISHABLE (Crosses Eq)
0.5c (0.5%)	395	199	50.4%	[45.5%, 55.3%]	50.5%	INDISTINGUISHABLE (Crosses Eq)
1.0c (1.0%)	355	190	53.5%	[48.3%, 58.6%]	51.1%	INDISTINGUISHABLE (Crosses Eq)
1.2c (1.2%)	337	176	52.2%	[46.9%, 57.5%]	51.1%	INDISTINGUISHABLE (Crosses Eq)
1.5c (1.5%)	313	160	51.1%	[45.6%, 56.6%]	50.2%	INDISTINGUISHABLE (Crosses Eq)
2.0c (2.0%)	264	141	53.4%	[47.4%, 59.3%]	50.7%	INDISTINGUISHABLE (Crosses Eq)
3.0c (3.0%)	169	98	58.0%	[50.5%, 65.2%]	49.8%	PASS (95% CI > Break-Even)
5.0c (5.0%)	72	41	56.9%	[45.4%, 67.7%]	51.8%	INDISTINGUISHABLE (Small Sample)

Table 2: Autonomous Market Maker Strategy Performance (405 Resolved Buckets in `pmbtc.db`)

Strategy	Total Fills	Gross Cumulative PnL	Execution Friction (18%)	Real Net PnL
Autonomous Market Maker	20,662 fills	+\$5,403.20 USD	-\$972.58 USD	+\$4,430.62 USD